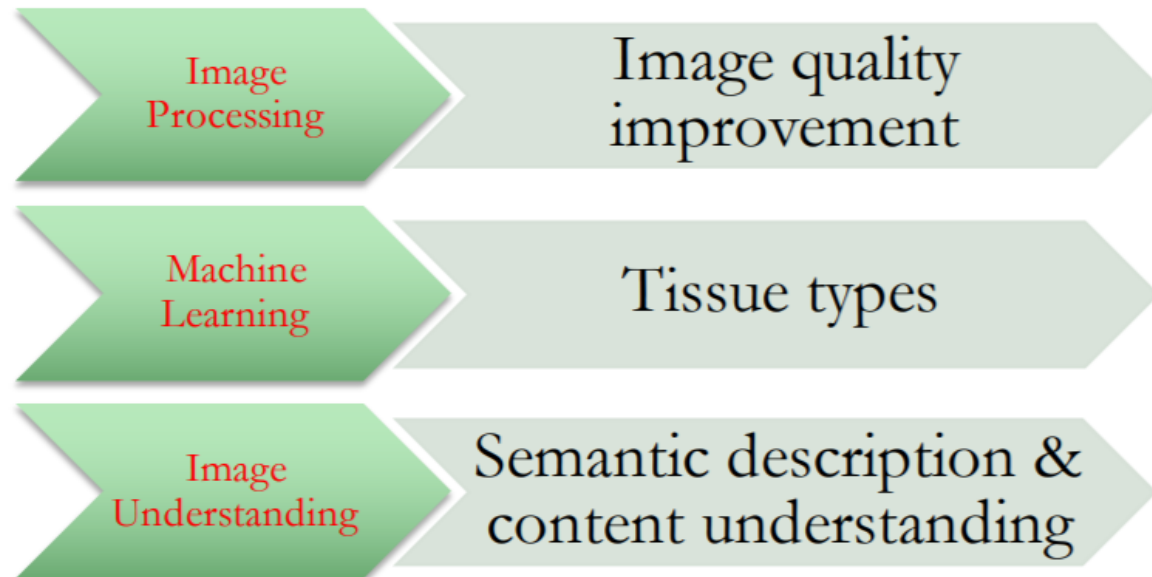


Multimodal Biomedical Image Analysis

Nada Fitrieyatul Hikmah

Medical Imaging

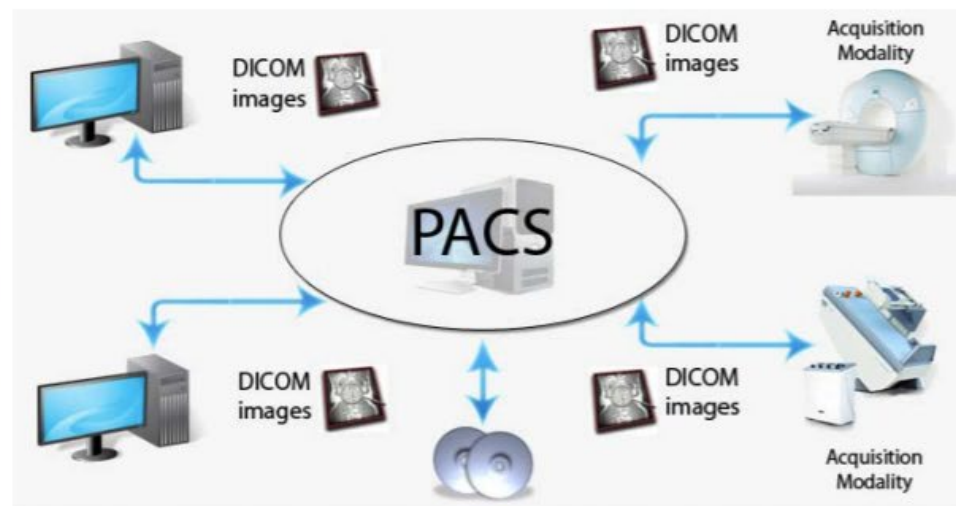
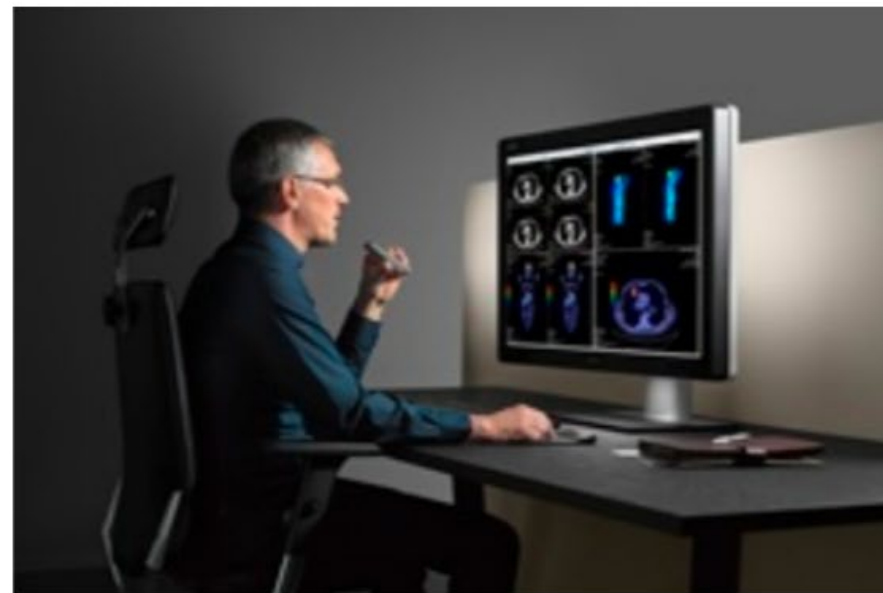
- The most direct way to see inside the human (or animal) body is cut it open (i.e., surgery)
- We can see inside the human body in ways that are **less invasive** or (completely **non-invasive**)
- We can even see metabolic/functional/molecular activities which are not visible to naked eye



Where do radiologists interpret scans?

- Dedicated light source
- Darkened environment
- Limited distraction

PACS (Picture Archiving and Communication System)





PACS System: Everything you need to know about them

Introduction of PACS

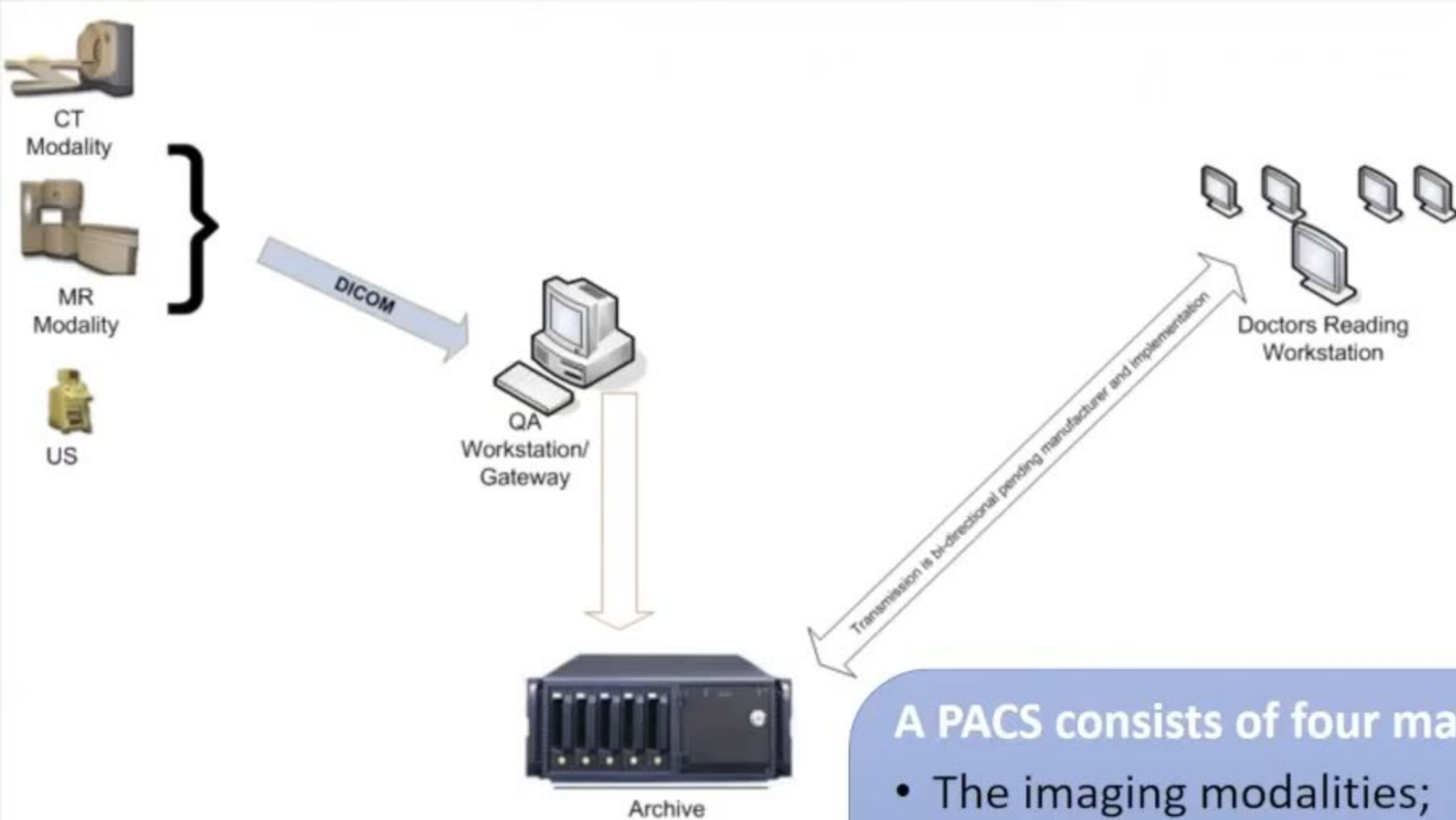
Picture archiving and communication system (PACS) has allowed the medical images to be transmitted, stored, retrieved, and displayed in different locations of a hospital or health system.





DICOM

The universal format for PACS image storage and transfer is DICOM (Digital Imaging and Communications in Medicine). DICOM permits PACSs, Radiology Information System (RIS) and more medical imaging systems to connect with and pass data to systems at other healthcare facilities.



A PACS consists of four major components:

- The imaging modalities;
- A secured network for the transmission of patient information;
- Workstations for interpreting and reviewing images;
- And archives for the storage and retrieval of images and reports.

Advantages

Hard copy replacement

PACS replaces hard-copy based means of managing medical images, such as film archives.



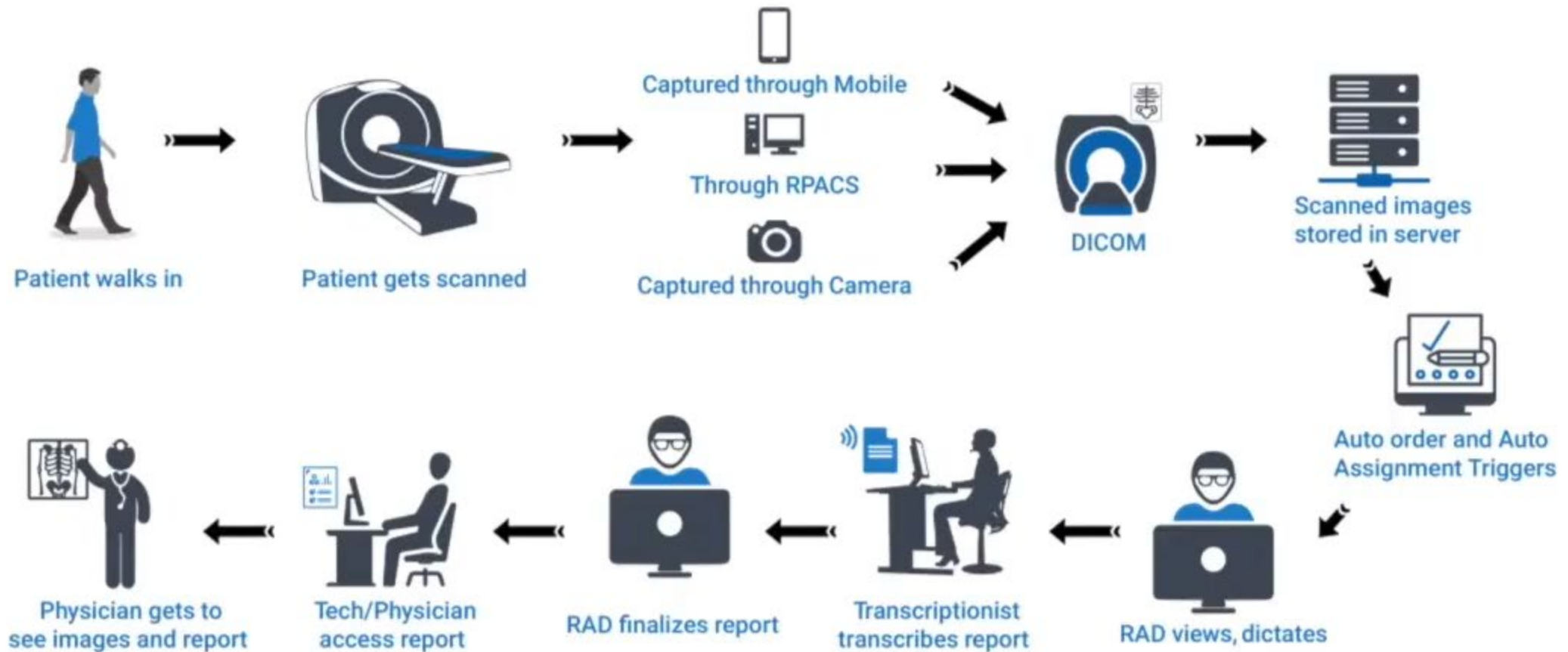


Remote access

It enables practitioners in different physical locations to access the same information simultaneously for teleradiology.

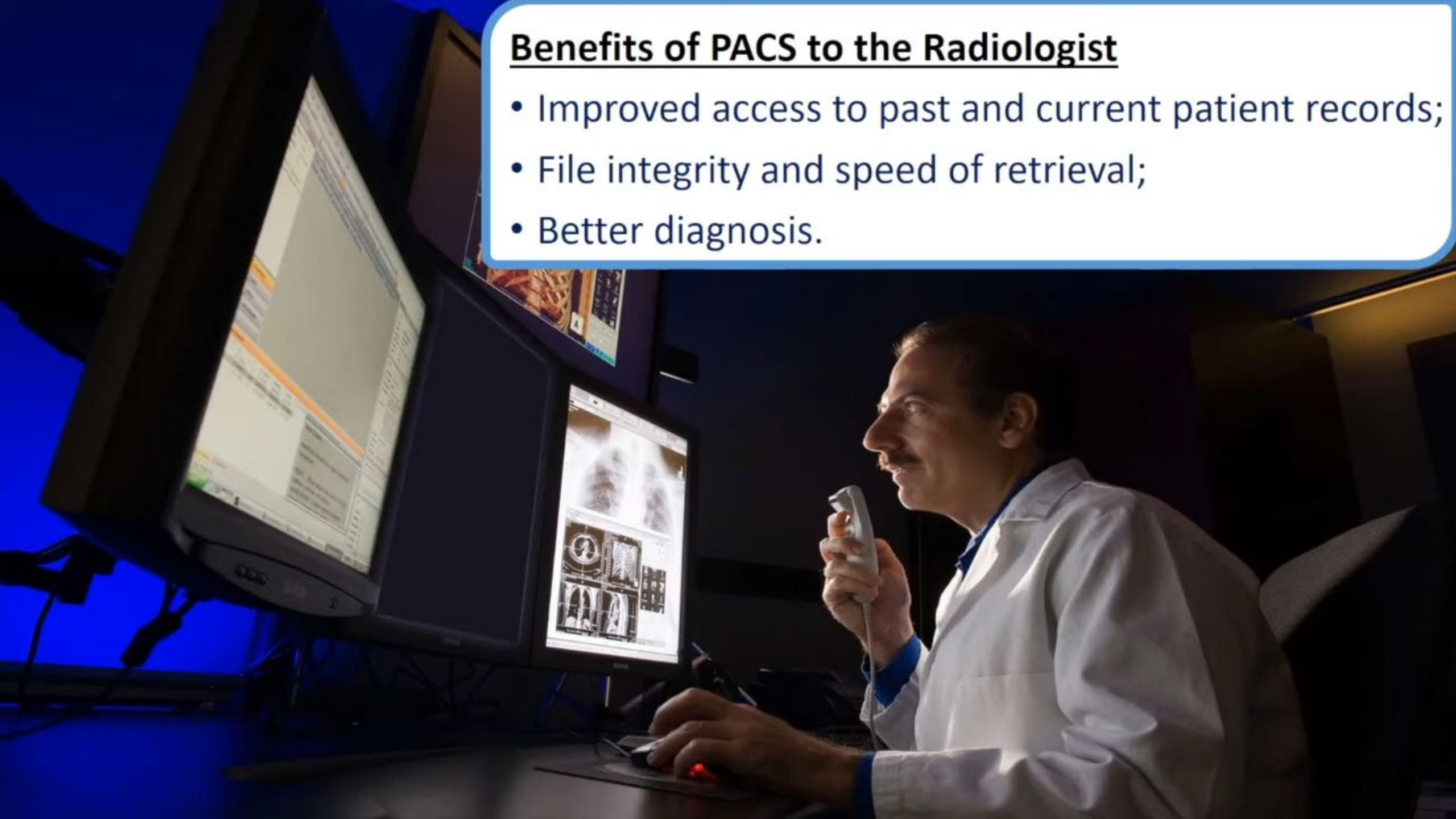
Radiology Workflow Management

PACS is used by radiology personnel to manage the workflow of patient exams.



Benefits of PACS to the Radiologist

- Improved access to past and current patient records;
- File integrity and speed of retrieval;
- Better diagnosis.





Benefits of PACS to the patient

- Shorter examination times;
- Reduced radiation exposure as a result of less need for retakes of images;
- Reduced patient inconvenience in attending hospitals for examinations and reexaminations .

Benefits of PACS to the hospital

- Better communication with physicians;
- Better hospital administration;
- Better training of radiology and other students through access to on-line image files and to digital teaching files;
- Greater staff retention due to improved morale.



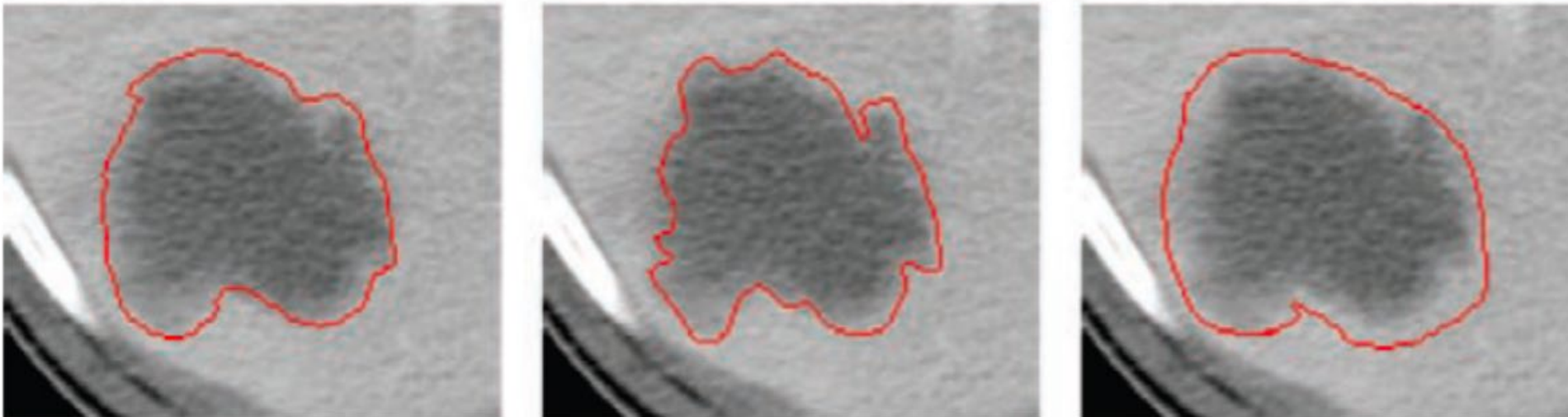
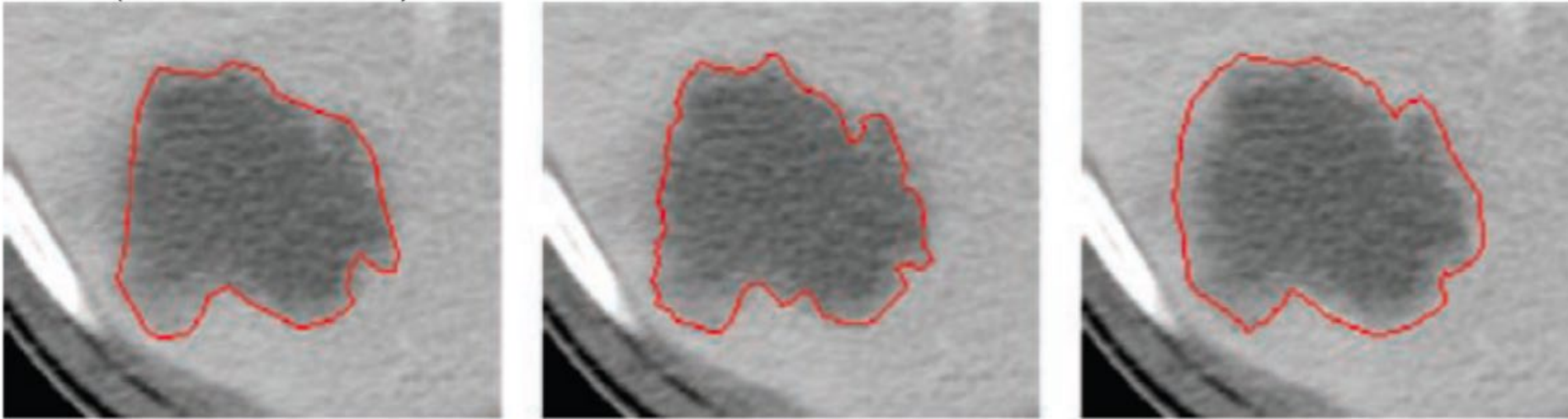


Medical Image Analysis - Manual

- Often accepted as surrogate of the truth (if biopsy or real ground truth is not available)
- However, manual analysis is highly subjective because it relies on the observer's perception.
 - Intra and inter-observer agreements/variabilities
- It is highly tedious

Observer Variability – Example: Liver lesion

Intra- (one week interval)

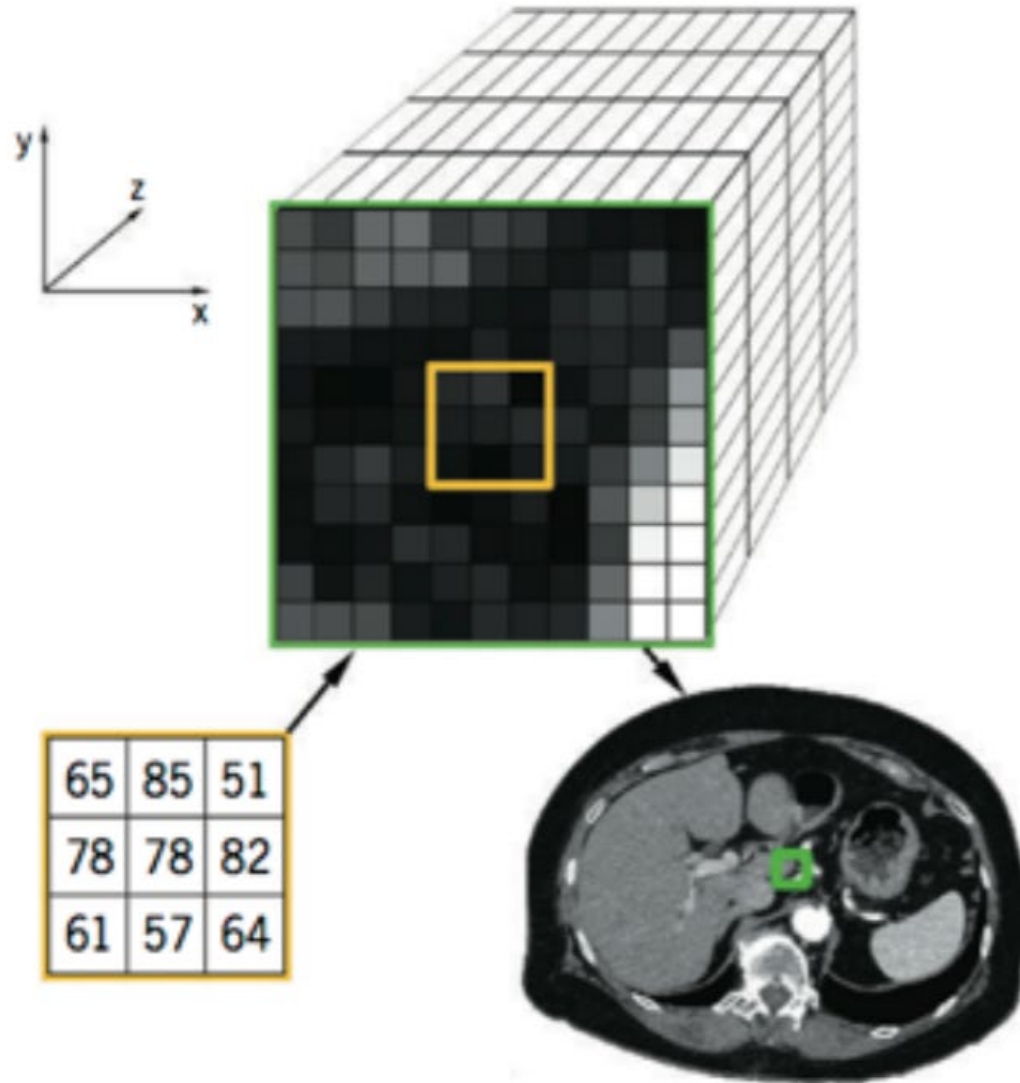


Inter-

Medical Image Analysis - Automated

- Different strategies for image analysis exist. However, few of them are suited for medical applications.
- The reason is that both the medical image data and the model or prototype (i.e., the a priori description of the features to be analyzed), are typically quite complex.

Digital Images

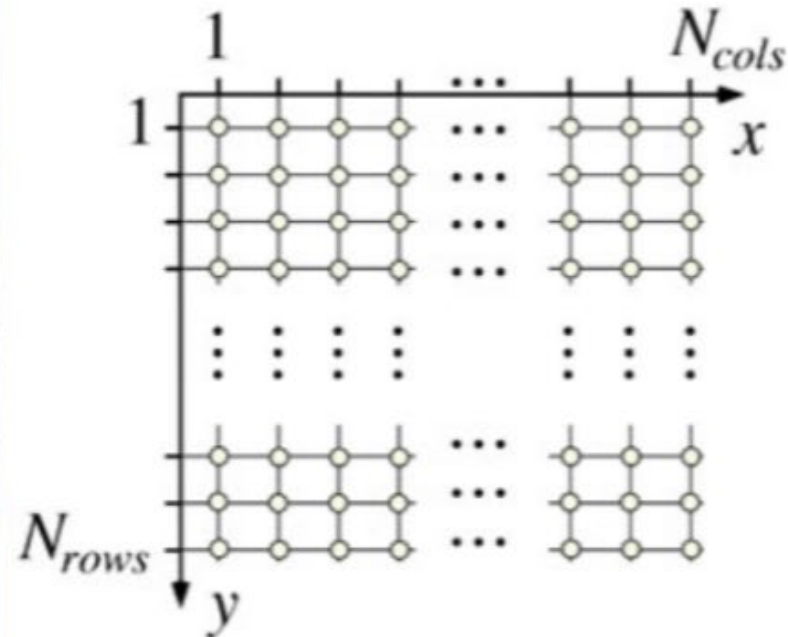


0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What computer sees!

Digital Images

- **Definition:** A digital image is defined by *integrating* and *sampling* continuous (analog) data in a spatial domain [Klette, 2014].



Picture Elements (Pixels), Volume Elements (Voxels)

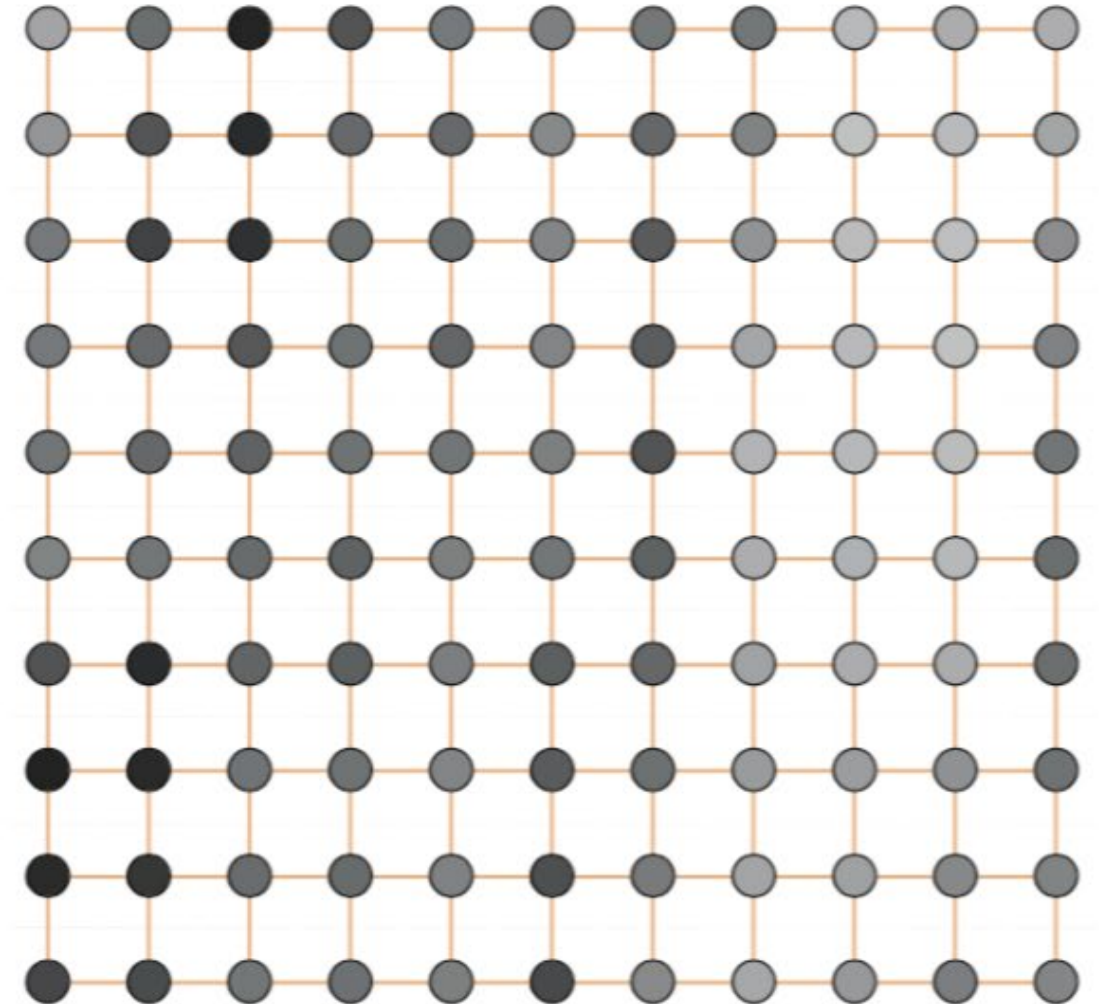
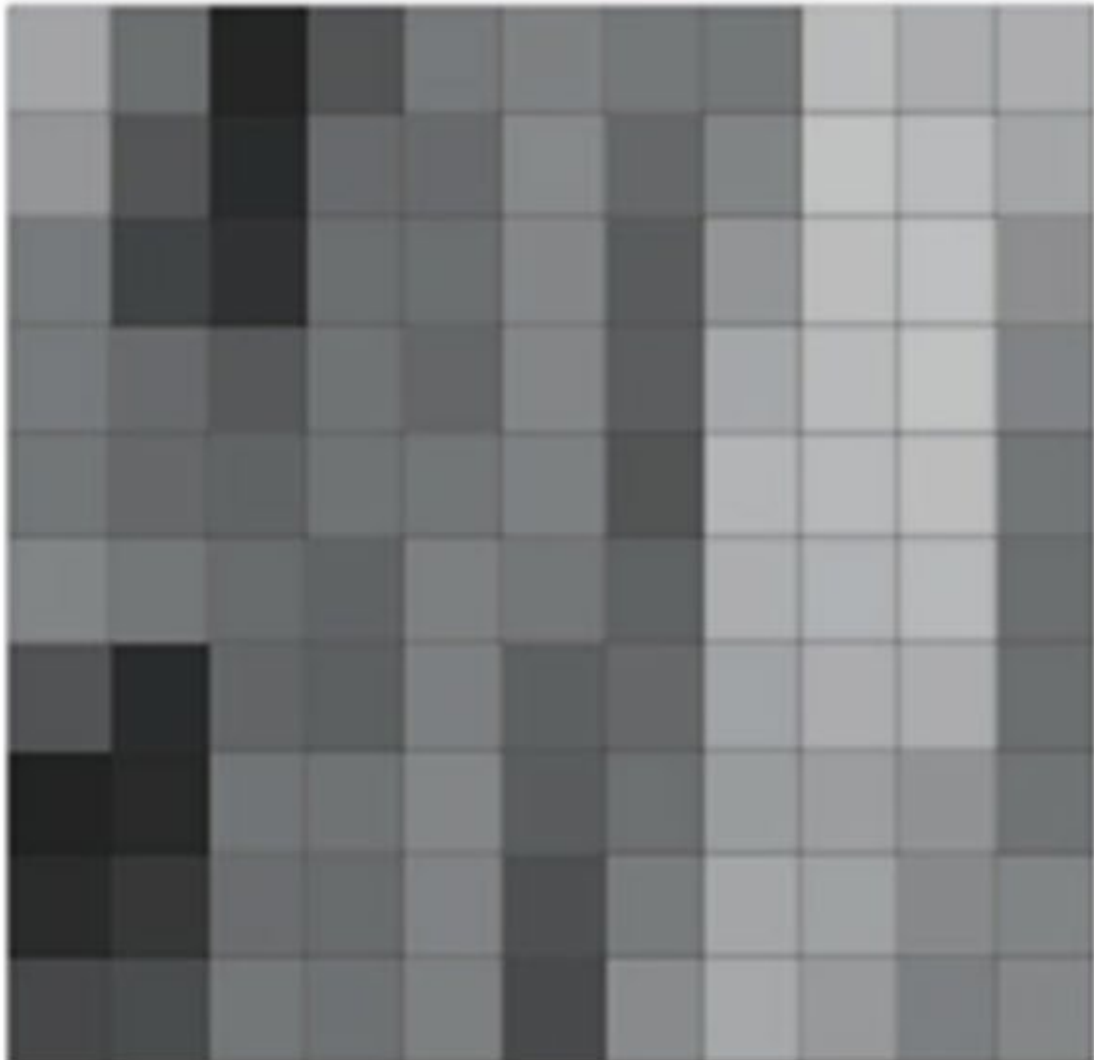


Image Types

- A scalar image has integer values

$$u \in \{0, 1, \dots, 2^a - 1\}$$

a: level (bit)

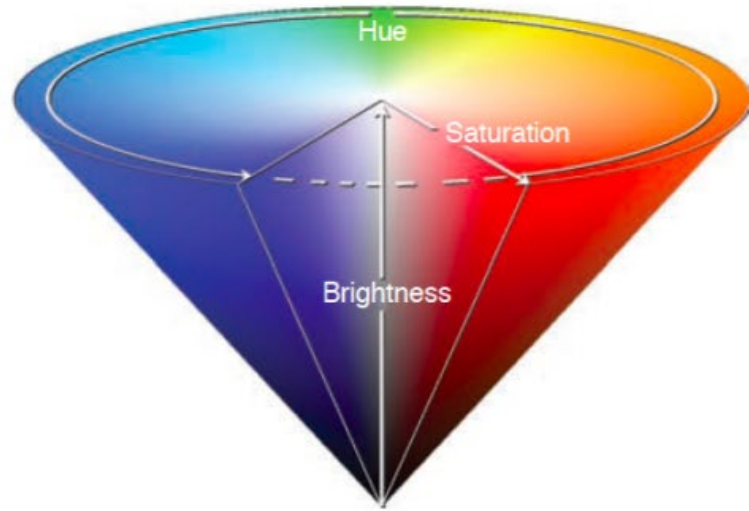
Ex. If 8 bit (a=8), image spans from 0 to 255

0 black

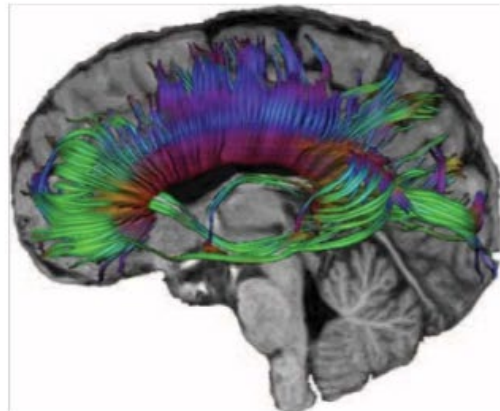
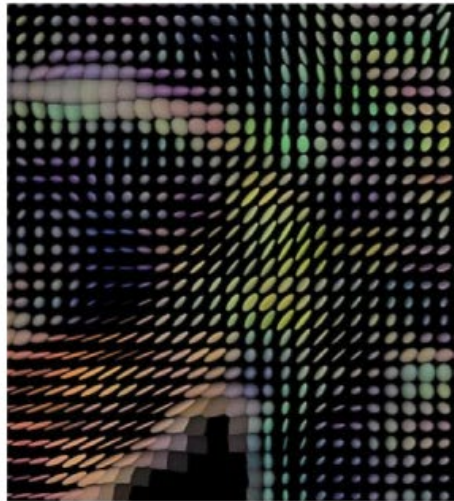
255 white

Ex. If 1 bit (a=1), it is binary image, 0 and 1 only.

Image Types - Color



- Image has three channels (bands), each channel spans a-bit values.
- RGB, Hue-Saturation-Brightness



A vertical bar on the left side of the slide, transitioning from orange at the top to purple at the bottom.

MULTIMODAL IMAGE

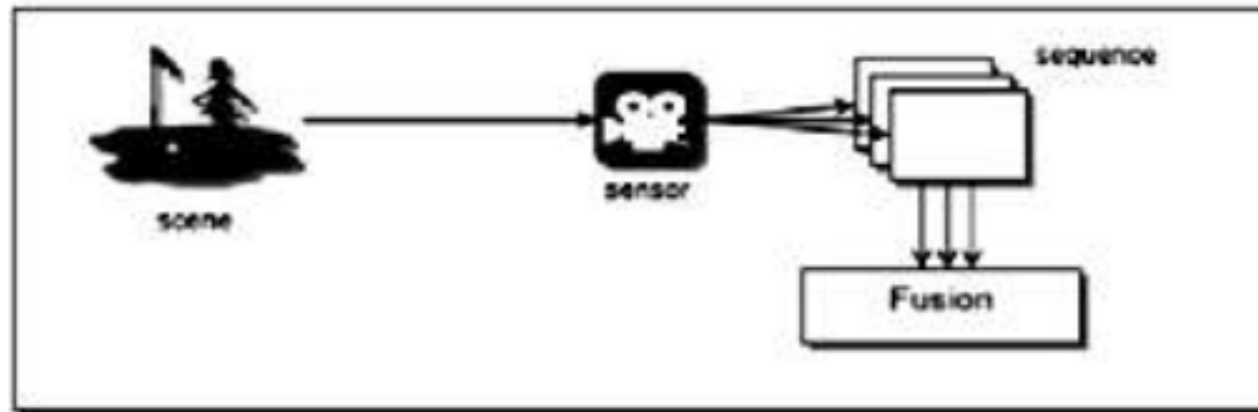
Multimodal Image

The objectives of multimodal image are to:

- Reduce amount of data → combining multiple images into a single image.
- Retain important information
- Create new image that is more suitable for the purposes of human/machine perception or for further processing tasks.

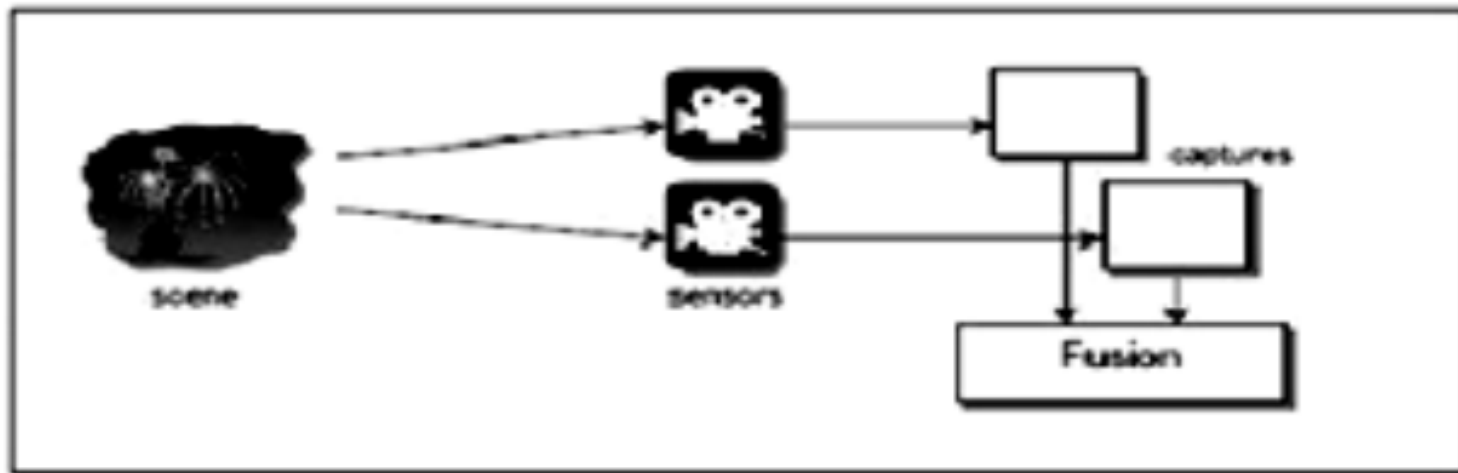
Single Sensor image fusion system

- Sequence of images are taken by the sensor
- Then they are fused in an image
- It has some limitations due to capability of sensor



Multi-sensor image fusion

- Images are taken by more than one sensor
- Then they are fused in an image
- It overcomes limitations of single sensor system



Fusion Categories

Multi-view fusion

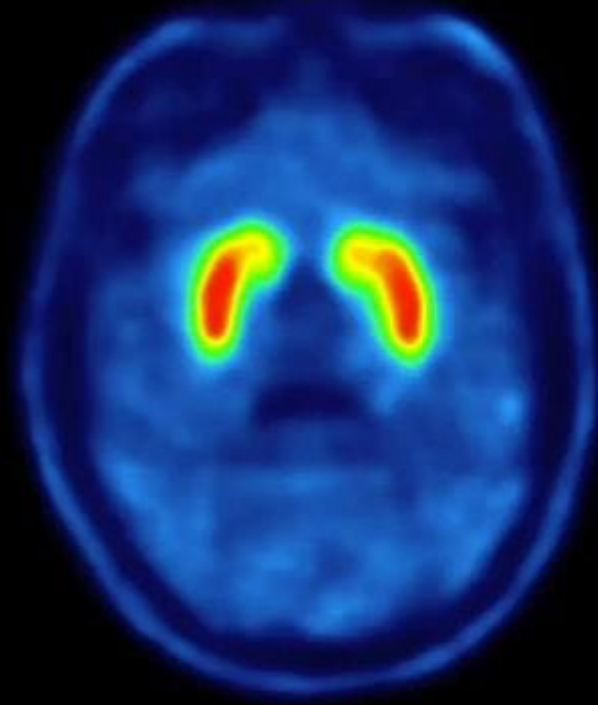
- Images are taken from different viewports to make 3D View
- Multi-modal fusion
- Multi-focus fusion

PET vs. MRI

Exceptional sensitivity & specificity
for a wide range of biological processes

Quantitative technique

PET

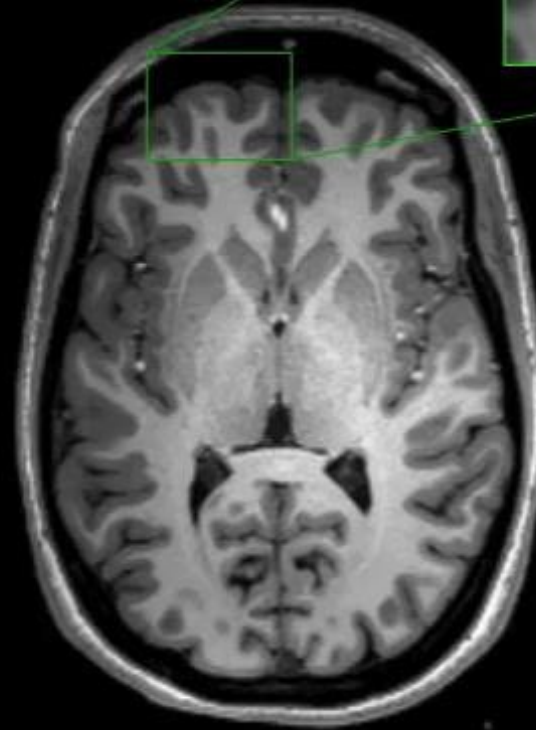


Lesser spatial resolution

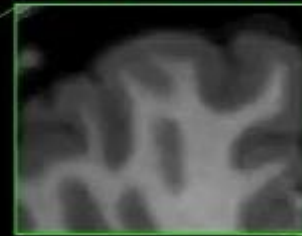
Limited anatomical information

Excellent
soft tissue
contrast

MRI



High resolution anatomy



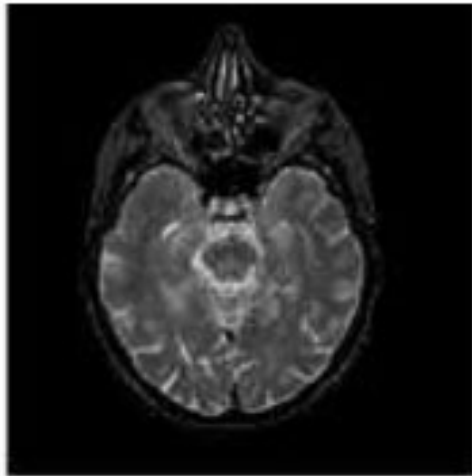
Good temporal
resolution

Poor sensitivity for molecular
processes

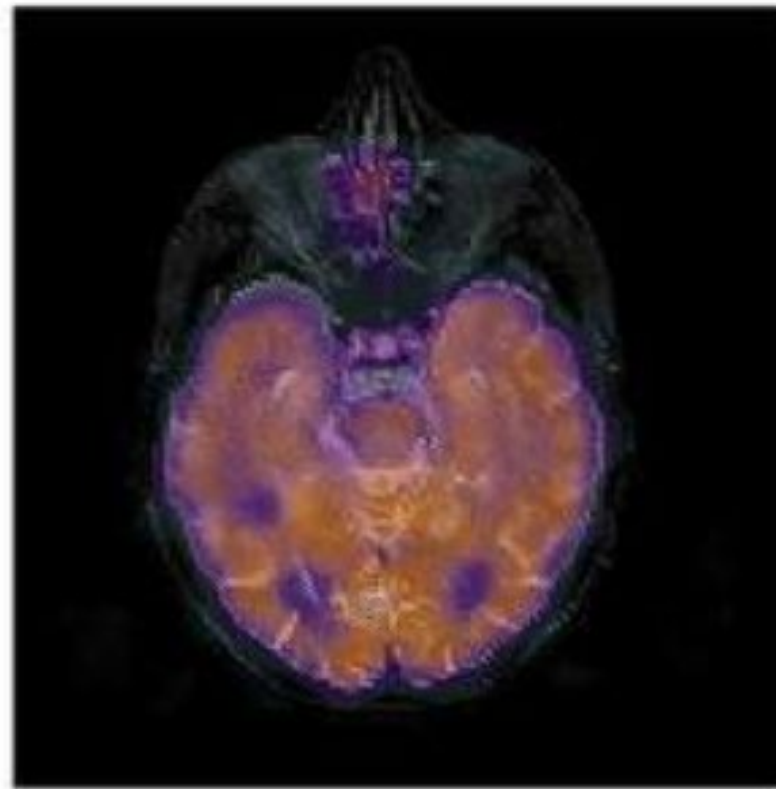
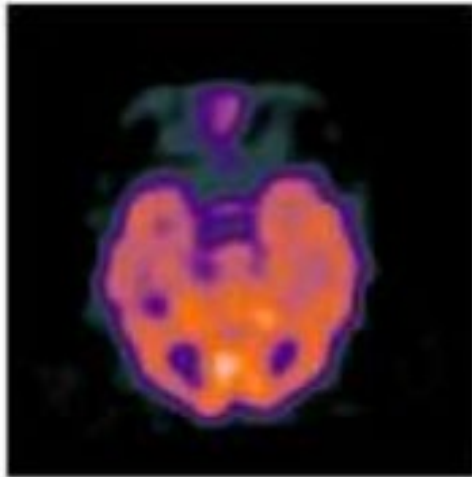
Multi-modal Fusion

Nuclear Magnetic Resonance Imaging

N
M
R



S
P
E
C
T



Multi-focus fusion



Fused image

A vertical bar on the left side of the slide, transitioning from orange at the top to purple at the bottom.

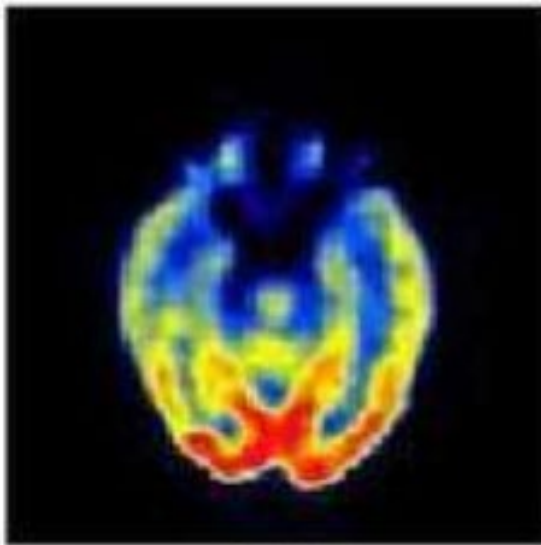
APPLICATIONS OF IMAGE FUSION

Medical Image Fusion

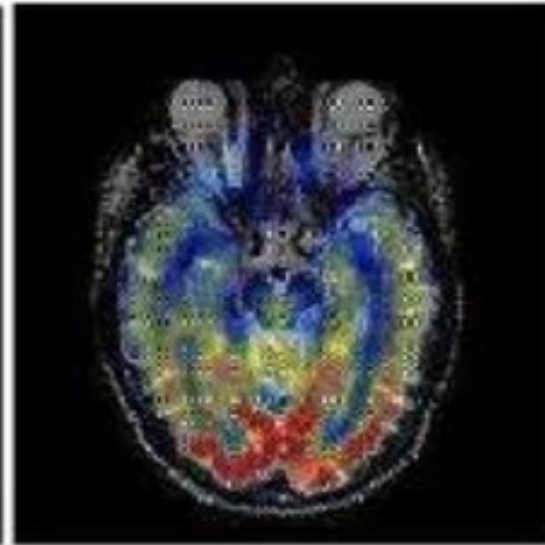
- Helps physicians to extract features from multi-modal images.
- Two types- structural (MRI, CT) & functional (PET, SPECT)



MRI-T2



PET



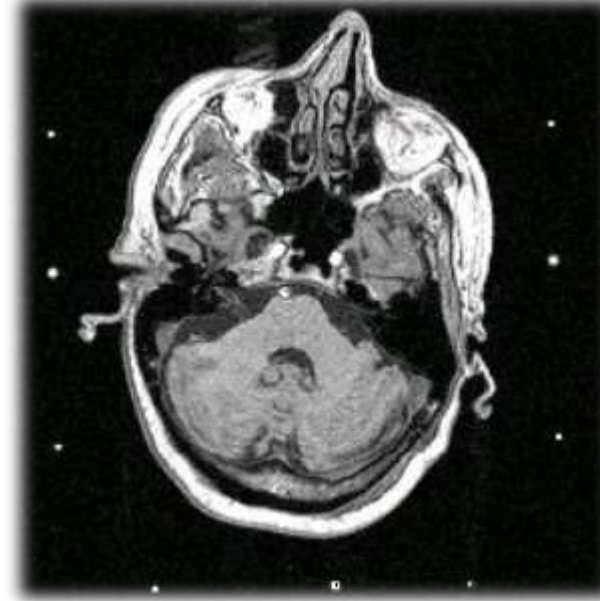
Fused

CT - MRI (1)

CT shows bony structures – very accurate



MRI image volume: soft tissue, show presence of a tumour



CT – MRI (2)



Fusing the two volumes, to find the most probable image is the core of automated radiation therapy planning

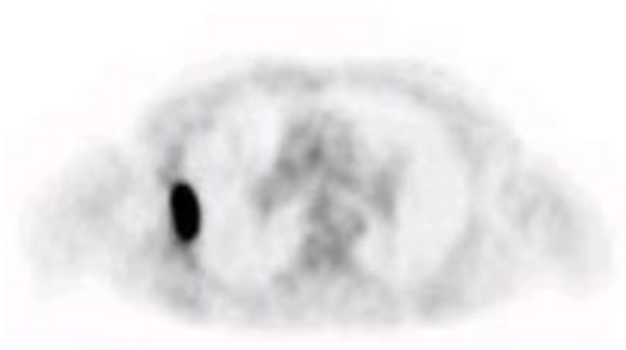
Advantages

- Better tumor delineation
- Spares normal tissue
- Better image quality

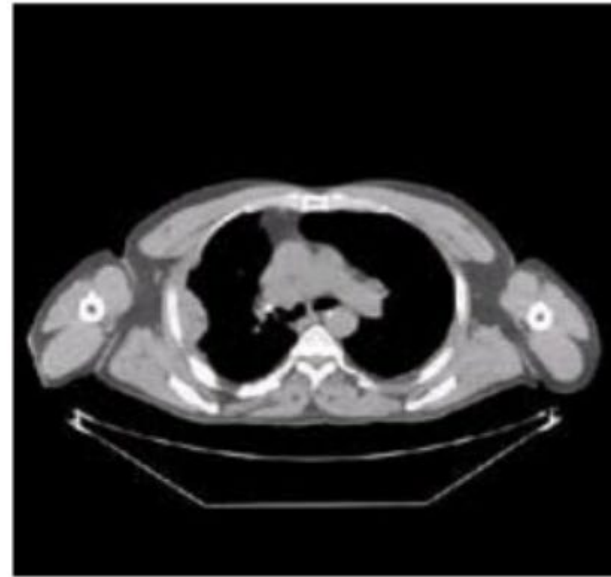
Disadvantages

- Expensive
- Time consuming

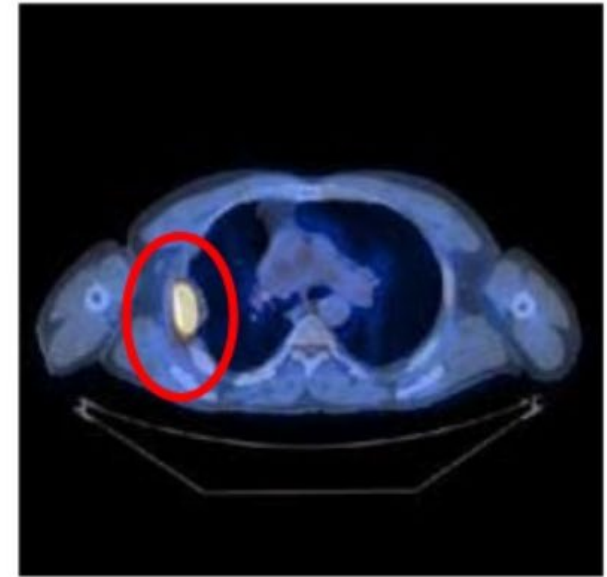
PET - CT



PET Image of the
Chest

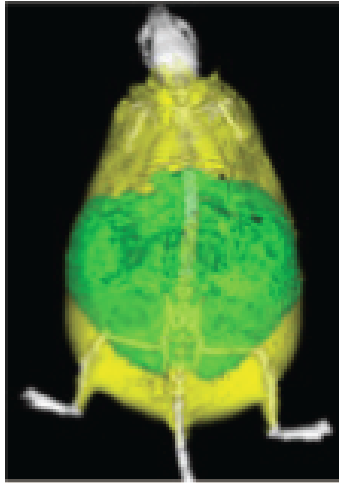


CT Image of the
Chest

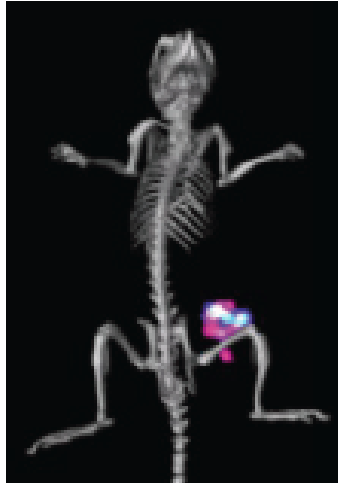


PET/CT Fused

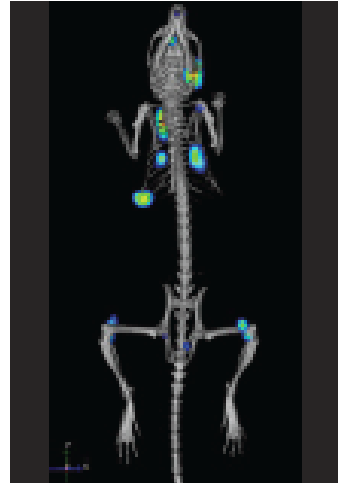
Optical and microCT



Metabolic disease

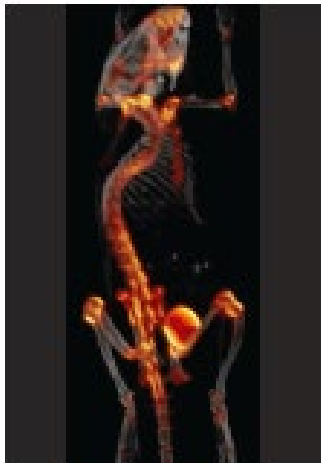


Cancer targeting

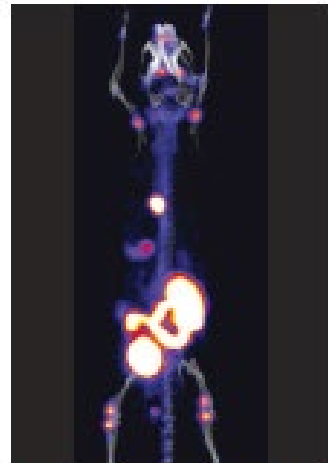


Multimodal imaging

PET/CT



Bone metabolism
(NaF-18)

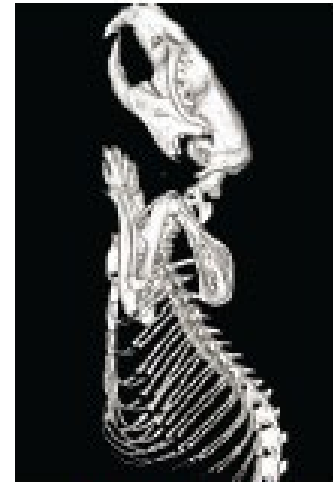


ImmunoPET with ^{89}Zr
labeled antibodies

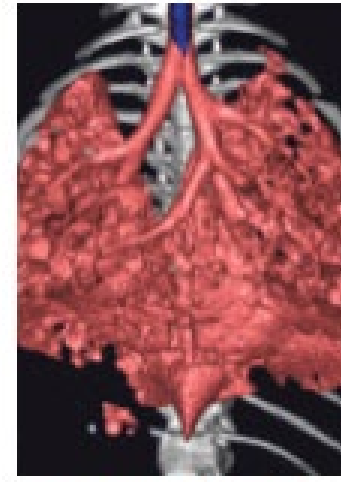


^{18}F -Fallypride imaging
of dopamine receptors in
a rat brain

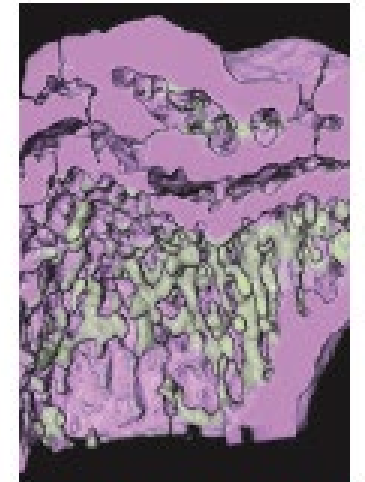
High Resolution microCT



Gross anatomy



Pulmonary disease



Trabecular bone imaging

Preparation

- Install Python
- Install 3D Slicer